

SAHAN VISHWA PERERA

SOFTWARE ENGINEER | TYPESCRIPT · WEBMCP · IOT | IT UNDERGRADUATE,
UNIVERSITY OF MORATUWA

☎ +94 75 615 6860 • ✉ svishwa0800@gmail.com • 📍 Werellagama, Sri Lanka • 🔗 linkedin.com/in/svishwa0800 • 🐙 github.com/Svishwa2004

PROFESSIONAL SUMMARY

Software-engineering undergraduate at the **University of Moratuwa** (B.Sc. Hons IT & Management, 2025–2029) building **privacy-preserving WebMCP agent tooling (TypeScript/Vite/Node.js)** and **production IoT firmware (C++/ESP32)**. Shipped **Airlock** — in-browser spreadsheet analysis an AI agent queries without receiving the file (live demo) — and **WebMCP-Gauge** — lint + run harness (4,500 trials) measuring whether agents call exposed tools. Also sole engineer of an automated optical pH sensor (12-state FSM, custom PCB, WebSocket dashboard). Seeking a **software-engineering internship**.

EDUCATION

B.Sc. (Hons) in Information Technology & Management

University of Moratuwa, Moratuwa

Expected Completion: 2029

2025 – Present

G.C.E. Advanced Level

St. Anthony's College, Kandy

ICT: A Accounting: A Economics: B

2023

G.C.E. Ordinary Level

St. Anthony's College, Kandy

IT: A English: A Religion: A + 6 more subjects

2020

WORK EXPERIENCE

IT Technician (Internship)

Kandy District Secretariat · Kandy, Sri Lanka

Jan 2022 – Jun 2022

- Maintained **desktop hardware, OS imaging, and software** for a government IT environment — installs, troubleshooting, and backups across multiple departments
- Streamlined **Excel/Word data-entry and document workflows**, improving record accuracy and retrieval; diagnosed connectivity and config issues to cut downtime

TECHNICAL SKILLS

PROGRAMMING LANGUAGES

C C++ Java JavaScript TypeScript Node.js SQL HTML5 CSS3

EMBEDDED SYSTEMS & IOT

ESP32 Microcontroller Programming Firmware Development Finite State Machine (FSM) PWM Control WebSocket

LittleFS PCB Design (EasyEDA) PlatformIO

WEB & AI-AGENT TOOLING

WebMCP Tools Vite PapaParse REST APIs Chrome CDP LLM Judge Evals GitHub Actions CLI Development

IT SUPPORT & SYSTEMS

Hardware Troubleshooting OS Configuration Network Connectivity Data Backup & Recovery

DEVELOPER TOOLS

Git GitHub VS Code Version Control Microsoft Office Suite

SOFT SKILLS

Analytical Problem Solving Technical Communication Self-directed Learning

 CERTIFICATIONS

National Certificate — ICT Technician (NVQ Level 4) 2022
College of Technology, Kandy

 PROJECTS

Airlock — Private In-Browser Spreadsheet Analyst (Devpost WebMCP Hackathon) — [GitHub](#) · [Live Demo](#)

TypeScript · Vite · WebMCP · PapaParse · HTML/CSS

- Built for the **Devpost WebMCP Hackathon challenge** — a **privacy-preserving data workbench** where CSV parsing and all aggregations run entirely in the browser tab — an AI agent queries 7 WebMCP tools (**describe_dataset, sum_by_category, filter_rows, monthly_trend, find_anomalies, top_expenses, clear_highlights**) and receives only aggregates while the network stays silent
- Designed a **default-on privacy gate** (file name + row details withheld, matches highlighted on the human's screen) with unit tests per tool; documented 5 empirically verified WebMCP API behaviors against Chrome 151

WebMCP-Gauge — Agent Tool-Invocation Measurement Harness — [GitHub](#)

Node.js 24 · JavaScript (ESM) · Chrome CDP · LLM Judge Evals · GitHub Actions

- Shipped **lint (static manifest rules) + run (S sessions × R repeats, Wilson intervals, JSONL resume, honest Shields badges)** CLI plus a **Marketplace GitHub Action** with split exit codes separating "could not measure" from genuine regressions
- Published **16 runs / 4,500 trials**: clean manifest **99.3%** vs deliberately degraded twin **83.1%** (sum_by_category 60.0%, top_expenses 26.7%), proving the metric discriminates real description defects

Automated Optical pH Sensor for Hydroponic Systems — [GitHub](#)

C++ · ESP32 · PlatformIO · FSM · WebSocket · LittleFS · PCB Design (EasyEDA) · HTML/CSS/JavaScript

- Sole engineer of a **production-grade automated IoT colorimetric pH sensor** — embedded C++ firmware, custom PCB, 3D-printed optics enclosure, 3-pump fluidics, and real-time web dashboard from concept to completion
- Architected a **12-state non-blocking FSM** (fill → dose → mix → Dynamic Blanking → measure → vacuum drain) with dual-wavelength optics, TCS34725 HSV colorimetry, and NVS calibration for reliable hands-free operation